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An interpretable delta ultrasound radiomics model for predicting live birth outcomes in single vitrified-warmed blastocyst transfer

Lidan Liu^{1*†}, Huimei Wu^{1†}, Qianyi Huang¹, Dongliu Xu², Mujun Li¹ and Lang Qin^{1*}

Abstract

Objective To develop and validate an interpretable delta ultrasound radiomics model for predicting live birth following single vitrified-warmed blastocyst transfer (SVBT).

Design Retrospective cohort study.

Setting The First Affiliated Hospital of Guangxi Medical University.

Patients 528 infertile women who underwent SVBT between January 2019 and December 2022.

Intervention Delta radiomics analysis of the gestational sac at 6th and 8th weeks' gestation, with extraction of radiomic features and integration with clinical parameters.

Main outcome measures Live birth, defined as the delivery of a single infant at ≥ 28 weeks of gestation. Radiomic feature differences between weeks 6 and 8 (delta features), combined with maternal age, were used to build ten machine learning models. The best-performing model was interpreted using SHapley Additive exPlanations (SHAP).

Results A delta radiomics model based on logistic regression yielded superior predictive performance compared to models using only single-timepoint features, achieving an AUC of 0.883 in the training cohort. Incorporating maternal age resulted in an AUC of 0.747 in the testing cohort. SHAP analysis confirmed that both delta radiomic features (from weeks 6 and 8) and maternal age substantially influenced model predictions.

Conclusion The Delta ultrasound radiomics model with SHAP-based interpretability offers a comprehensive and dynamic approach for predicting live birth in SVBT.

Trial registration Not applicable.

Capsule A delta ultrasound radiomics model incorporating 6 and 8 week gestational sac features, combined with clinical factors, improves live birth prediction following SVBT.

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Keywords Delta radiomics, Live birth prediction, Single vitrified-warmed blastocyst transfer(SVBT), SHAP analysis, Gestational sac

Introduction

Infertility is a significant global health issue, affecting approximately one in six couples worldwide, according to the World Health Organization (WHO) in 2023 [1]. This striking prevalence not only highlights the extensive scale of the challenge but also underscores the growing imperative for advanced and accessible reproductive healthcare. Assisted reproductive technology (ART) has provided hope to countless couples experiencing infertility [2]. Among various ART methods, single vitrified-warmed blastocyst transfer (SVBT) is gaining popularity, as it improves embryo selection and reduces the risks associated with multiple pregnancies and related complications [3, 4]. Blastocysts undergo a more rigorous natural selection process compared to cleavage-stage embryos, effectively eliminating those with lower developmental potential [5, 6]. Despite these advantages, live birth rates following SVBT remain inconsistent, and predicting successful pregnancy outcomes continues to pose a significant clinical challenge [2]. Accurate prediction of live birth outcomes is crucial for improving treatment strategies and providing effective patient counseling, particularly during the first trimester [7]. Therefore, developing more precise and customized prediction models for live birth outcomes in early pregnancy is essential.

Radiomics, which involves the extraction and analysis of quantitative features from medical imaging, offers a promising approach for predicting clinical outcomes [8]. In ART, ultrasound imaging, a widely used and non-invasive modality, holds great promise for predicting early pregnancy outcomes. In particular, ultrasound at the sixth week of pregnancy shows the gestational sac clearly, along with the yolk sac and fetal heartbeat, signifying successful embryo implantation. By the eighth week, the embryo continues to develop within the uterus, and the development of the gestational sac directly reflects the health of the pregnancy. Abnormalities in the development of the gestational sac at this stage may indicate potential complications such as miscarriage. These ultrasound images provide crucial early indicators for clinical intervention. The changes observed in the gestational sac between the sixth and eighth weeks offer dynamic imaging data, enabling more comprehensive and accurate evaluation of pregnancy progress. By comparing imaging features from different time points, the delta radiomics approach enhances predictive accuracy and provides a more robust model for early pregnancy outcomes.

Combining radiomic features from ultrasound images with clinical data may improve the accuracy of pregnancy outcome predictions. One promising technique, delta

radiomics, integrates features from ultrasound images taken at different times, potentially enhancing predictive models [9]. Delta radiomics technology has been explored for predicting radiation-induced esophagitis, pneumonitis, and other diseases, and has demonstrated promising predictive value [10–13]. By integrating imaging data from various time points, the delta radiomics approach may provide a comprehensive understanding of the early pregnancy course, circumventing the limitations of static, single-time-point prediction models.

However, to date, there has been no published research exploring the application of delta ultrasound radiomics in the field of ART. Therefore, our study aims to develop and validate one robust model for predicting live birth outcomes in women undergoing SVBT based on delta radiomics model. The model was constructed by integrating ultrasound-derived radiomic features from sixth week and eighth week gestational scans, and clinical features such as maternal age, fertilization method, and endometrial preparation protocols. To improve the interpretability of the delta radiomic model, Shapley Additive Explanations (SHAP) were used to weight feature contributions [14]. The findings of our study are intended to offer a valuable tool for clinicians to improve patient counseling and decision-making in ART.

This approach hypothesizes that by integrating imaging features from sixth week and eighth week gestational scans, delta radiomics could offer enhanced predictive power compared to traditional methods, and the application of SHAP is expected to improve the model's transparency and interpretability, thereby further augmenting the predictive performance of delta radiomics in forecasting early pregnancy outcomes.

Materials and methods

Study design and population

Women who received SVBT treatment at the first affiliated hospital of Guangxi Medical University in Nanning, China, between January 2019 and December 2022 were examined in this retrospective cohort study. We utilized ultrasound-derived radiomic features and clinical variables, with a primary focus on predicting live birth outcomes. The cohort design was chosen due to the availability of retrospective clinical data, which enabled us to evaluate the predictive potential of delta radiomics in early pregnancy without the need for intervention or randomization, as would be required in a clinical trial. Eligible participants were those with confirmed intrauterine pregnancies, as determined by ultrasound examinations conducted at sixth and eighth weeks of gestation (the

fourth and sixth weeks following transplantation). The exclusion criteria included: [1] biochemical pregnancy loss (defined as a positive β -HCG result without ultrasound confirmation of the gestational sac) [2], ectopic pregnancy or heterotopic pregnancy [3], multiple pregnancies [4], empty gestational sac [5], yolk sac without an embryo [6], absence of fetal heart activity [7], missing or low-quality ultrasound data for pregnancies at sixth or eighth weeks. The analysis included 528 cases in total that satisfied the inclusion criteria. The first affiliated hospital of Guangxi Medical University Ethics Committee gave its approval to the study on July 14, 2025 (approval number: 2025-E0580). The Ethics Committee waived the requirement for informed consent because the study was retrospective in nature and used early pregnancy ultrasound images and archived medical records. To preserve privacy, all patient data was anonymized.

Embryo culture and transfer

Each patient's ovarian stimulation protocol was individualized based on clinical factors such as age, BMI, baseline FSH levels, and antral follicle count. Ovulation was triggered with human chorionic gonadotropin (hCG) when at least one follicle reached a diameter of 18 mm, followed by oocyte retrieval 36 hours later under ultrasound guidance. The insemination method—either conventional in vitro fertilization (IVF) or intracytoplasmic sperm injection (ICSI)—was selected according to semen quality. Following fertilization, embryos were cultured continuously in a single-step medium under oil at 37°C, in an incubator environment containing 5% oxygen, 6% carbon dioxide, and nitrogen as the balancing gas. Blastocyst quality was assessed using the Gardner scoring system. Fully expanded blastocysts were laser-shrunk prior to vitrification using commercial vitrification kits (KITAZATO), loaded onto cryotops on day 5–6 post-insemination, and stored in liquid nitrogen. Thawing was performed using warming kits (KITAZATO) once the endometrial lining reached a sufficient thickness, and embryo viability was assessed based on re-expansion two hours post-warming. For frozen embryo transfer (FET), endometrial preparation followed one of four protocols—modified natural cycle (mNC), mild stimulation cycle (MS), hormone replacement therapy (HRT), or HRT combined with a GnRH agonist (GnRHa-HRT)—selected according to the patient's individual clinical profile. The endometrial preparation protocol was detailed in our prior publication (see 'Endometrial Preparation and Blastocyst Transfer' section) [15]. All blastocyst transfers were performed under abdominal ultrasound guidance to ensure accurate placement.

Clinical data and analysis

The primary clinical outcome of this study was live birth, defined as the delivery of a single infant at ≥ 28 weeks of gestation. Clinical data were extracted from the electronic medical records of participating centers to ensure accuracy. All key variables were independently verified by dual clinical data managers, including patient demographics and clinical data, which were derived from the medical records of patients diagnosed with a singleton intrauterine pregnancy following SVBT and ultrasound assessment at 4 weeks post-procedure. The retrospective analysis utilized clinical data reflecting known significant variables that may influence fetal development and pregnancy outcomes. The inclusion criteria for variables were based on established clinical knowledge and previous studies in the field of ART, with particular attention to factors known to influence pregnancy outcomes. These variables included maternal age, fertilization method (IVF or ICSI), duration of infertility, body mass index (BMI), embryonic development stage (assessed using the Gardner scoring system [16], quality grades of the inner cell mass (ICM) and trophectoderm (TE) [17], blastocyst formation time, endometrial thickness (EMT), and endometrial preparation protocols (including modified natural cycle, mild stimulation cycle, hormone replacement therapy, and GnRH agonist combined with HRT). These clinical data, which may significantly impact pregnancy outcomes, were carefully considered during the data analysis process.

Univariate Analysis: Clinical variables, including maternal age, fertilization method (IVF or ICSI), and endometrial thickness, were assessed using Mann-Whitney U tests for continuous variables and Chi-square tests for categorical variables. A p-value of less than 0.05 was considered statistically significant, and variables with a p-value < 0.05 were included in the subsequent multivariate analysis.

Multivariate Analysis: In the multivariate analysis, we used logistic regression to evaluate the independent effects of the selected variables on live birth outcomes. The final set of variables included those with a p-value < 0.05 in the univariate analysis. The odds ratios (OR) and corresponding 95% confidence intervals (CI) were computed for each variable, and the most significant variables, including endometrial thickness, were combined with the delta radiomics features to construct the final predictive model.

Ultrasonography protocol

All ultrasound examinations at sixth and eighth weeks of gestation were performed by experienced sonographers following standardized protocols. The TK-SNAP software (version 3.8.0) was used to manually delineate the regions of interest (ROIs) and annotate the gestational

sac. For the 528 participants, the initial ROI annotations were performed by Sonographer A. To rigorously assess intra- and inter-observer consistency of the extracted features and ensure methodological reproducibility, a random subset of 200 cases was re-annotated after a two-week interval. This re-analysis was independently conducted by Sonographer A (for intra-observer assessment) and Sonographer B (for inter-observer assessment), strictly adhering to the original annotation criteria. The aim of this process was to validate the robustness of feature selection and minimize variability in ultrasound-derived measurements.

Radiomic feature extraction and fusion

The open-source Python package PyRadiomics on the OnekeyAI platform was used to extract radiomic features of the gestational sac from ultrasound images taken at 6 and 8 weeks of gestation. In addition to morphological features, the extracted features comprised first-order statistical features and those obtained from multiple matrices, including the Gray Level Co-occurrence Matrix (GLCM), Gray Level Dependence Matrix (GLDM), Gray Level Run Length Matrix (GLRLM), Gray Level Size Zone Matrix (GLSZM), and Neighboring Gray Tone Difference Matrix (NGTDM). Following extraction, concatenation was used to perform feature fusion, which combined the features of the 6-week with 8-week datasets. This method preserved the independent data from each time point while producing a more complete and trustworthy representation of features for additional analysis and modeling.

Feature selection

Prior to statistical analysis and feature selection using the Mann-Whitney U test, radiomic features were first standardized using Z-score normalization. Only features with a p-value less than 0.05 were kept. Spearman's rank correlation coefficient was also used to evaluate inter-feature correlations for highly redundant features. Only one feature per pair of highly correlated features remained after features with a correlation coefficient greater than 0.9 were removed. The feature set was then optimized using logistic regression with the Least Absolute Shrinkage and Selection Operator (LASSO). 5-fold cross-validation was used to determine the optimal regularization parameter,

λ , which minimized binomial deviance. The selected features were used to construct a machine learning model, and the weighted linear combination of the selected features was used to calculate the radiomic score for each patient. Models created using each modality separately as well as a combined model that included all modalities had their classification performance assessed. The selected features were used to construct a machine learning model, and the features were then combined in a weighted linear fashion to determine each patient's radiomic score. The classification performance of models developed using each modality independently and a combined model incorporating all modalities was evaluated. A flowchart illustrating radiomics feature extraction, clinical feature selection, and model development is presented in Fig. 1.

Signature building

Delta Radiomics Signature Following a comprehensive feature selection procedure using LASSO, a radiomics-based live birth prediction model was developed using machine learning algorithms like Random Forest, Support Vector Machine, and Logistic Regression. Comparative analysis was used to evaluate each model's performance. Integrating features from the sixth and eighth weeks to examine the advantages of Delta temporal fusion further illustrated the advantages of combining imaging features from different time intervals to increase predictive accuracy.

Ultrasound Radiomics Signatures at Sixth and Eighth Weeks: Using a methodology akin to Delta radiomics, features were taken from ultrasound pictures taken at sixth and eighth weeks. The sixth and eighth week radiomics signatures were derived by selecting and reducing the dimensionality of these features. The efficacy of the Delta approach in boosting model robustness and predictive power was confirmed by contrasting these signatures with the Delta model.

Metrics

To assess our machine learning model's diagnostic performance in the test population and ascertain its discriminatory power, we developed Receiver Operating Characteristic (ROC) curves. The Hosmer-Lemeshow goodness-of-fit test was used to verify the model's

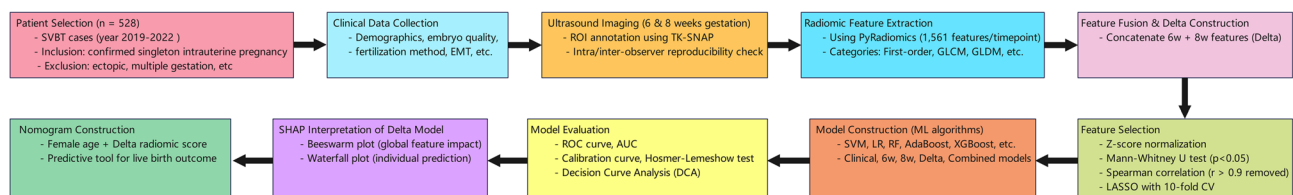


Fig. 1 Workflow of feature integration and predictive modeling for live birth outcome prediction

reliability after it had been calibrated using calibration curves. Additionally, Decision Curve Analysis (DCA) was used to assess the clinical utility of our predictive models, yielding valuable insights regarding their potential advantages in a clinical context. The nomogram was developed based on the relationship between these variables and the live birth outcome.

Model interpretation

To enhance the interpretability of the delta radiomics model, feature contributions were analyzed using SHAP method. Based on cooperative game theory, SHAP addresses the interpretability issues with traditional “black-box” models and improves the decision-making process by quantifying the marginal contribution of each feature to the model's predictions.

To assess the global impact of individual features on the overall live birth predictions, a Beeswarm plot was made, which displays the distribution of SHAP values for each feature across all samples [18]. For a more in-depth analysis of the prediction process for individual samples, a Waterfall plot was developed [19], which illustrates how the contributions of different features progressively impact the final live birth prediction outcome.

Statistical analysis

The Shapiro-Wilk test was used to determine whether the clinical features were normal. Depending on their distributional characteristics, continuous variables were analyzed using either the Mann-Whitney U test or the t-test. Chi-square (χ^2) tests were used to assess categorical variables. Notably, p-values greater than 0.05 between cohorts indicated no significant differences, confirming an unbiased group division. All data analyses were performed on the OnekeyAI platform (version 4.9.1) using Python 3.7.12. Statistical analyses were conducted using Statsmodels (version 0.13.2). Radiomics feature extraction was carried out with PyRadiomics (version 3.0.1). Machine learning models, including the Support Vector Machine (SVM), were implemented using Scikit-learn (version 1.0.2).

Results

Clinical characteristics of participants

The study included 528 patients in total, and Fig. 2 describes the recruitment procedure. The data was randomly split into training and testing groups with a 0.7:0.3 ratio. Clinical variables such as blastulation time, embryo stage, ICM, TE quality, maternal age at FET, BMI, duration of infertility, fertilization method, endometrial preparation protocol, and endometrial thickness at embryo transfer did not differ statistically significantly between the training and testing cohorts (Table 1). These results suggest that the clinical characteristics

of the training and testing groups were well matched, which served as a strong foundation for the subsequent analyses.

Sixth week ultrasound radiomic feature screening and model construction

1,561 ultrasound radiomic features including seven categories: First-order statistics, GLCM, GLDM, GLRLM, GLSZM, NGTDM and Shape features, were obtained by extracting the gestational sac's radiomic analysis. After Z-score normalization, Mann-Whitney U test, Spearman rank correlation analysis, and LASSO regression optimization, the optimal regularization parameter λ was 0.0146 (Fig. 3A). Ten statistically significant and mutually independent radiomic features were finally identified using LASSO regression, which successfully improved the feature ultrasound selection process at this regularization strength (Fig. 3B). Support Vector Machine (SVM), Logistic Regression (LR), Random Forest (RF), ExtraTrees, Naive Bayes (NB), XGBoost, Multilayer Perceptron (MLP), Gradient Boosting Machine (GBM), and AdaBoost were among the ten machine learning models whose performance was further compared using ROC curves. With an Area Under the Curve (AUC) of 0.663 in the testing group, the LR classifier outperformed the other models in terms of prediction, as seen in Fig. 3C.

Eighth week radiomic feature screening and model construction

As in the sixth week, 1,561 ultrasound radiomic features were extracted from the gestational sac. Following Z-score normalization, Mann-Whitney U test, Spearman rank correlation analysis, and LASSO regression, the optimal regularization parameter λ was found to be 0.0110 (Fig. 4A). LASSO regression identified sixteen statistically significant and independent features, enhancing the feature selection process at this regularization strength (Fig. 4B). To assess model performance, ten machine learning models were compared using ROC curves. Among these, XGBoost and LR outperformed the other models, achieving AUC values of 0.606 and 0.592, respectively, in the testing group, as shown in Fig. 4C.

Delta radiomic feature screening and model construction

Delta radiomic features represent the fusion of ultrasound imaging characteristics of the gestational sac from the sixth and eighth weeks. 3122 radiomic features were concatenated. Following Z-score normalization, Mann-Whitney U test, Spearman rank correlation analysis, and LASSO regression, the optimal regularization parameter λ was found to be 0.0126 (Fig. 5A). LASSO regression identified twenty-five statistically significant and independent features, enhancing the feature selection process at this regularization strength (Fig. 5B). To assess model

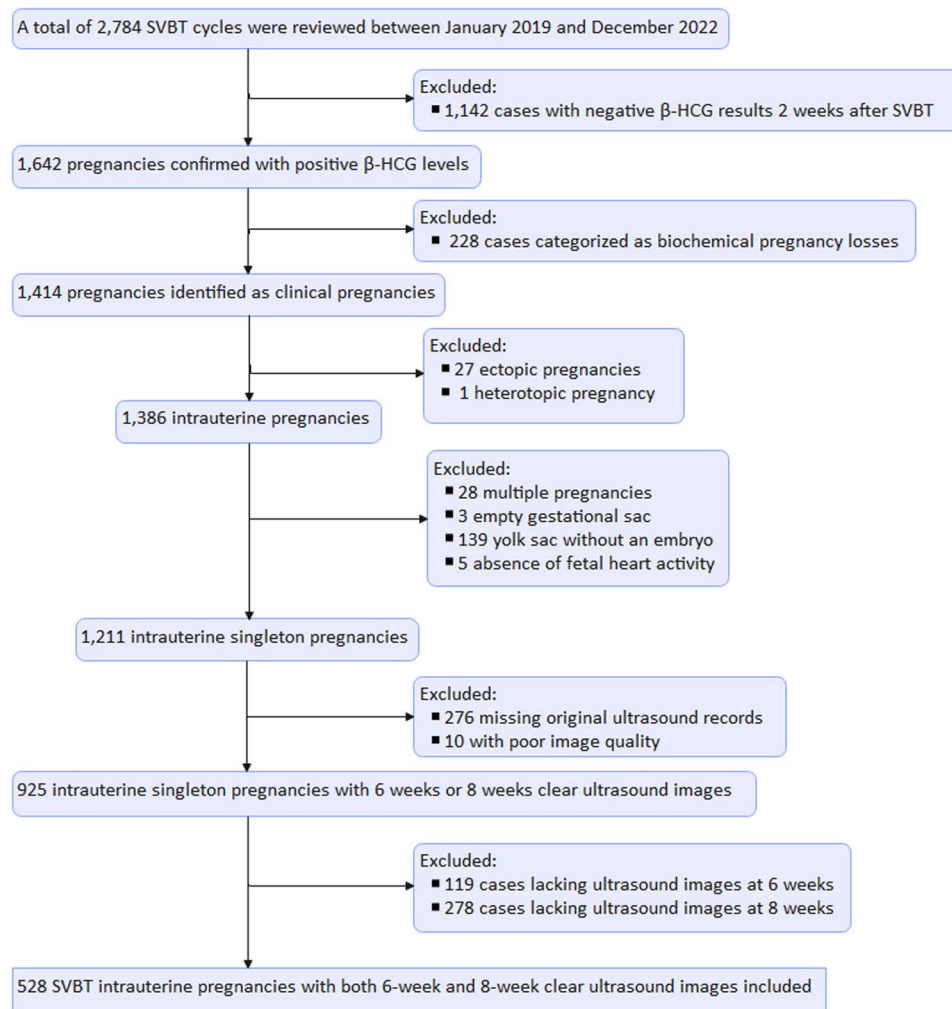


Fig. 2 Workflow of the study's data handling process

performance, ten machine learning models were compared using ROC curves. Among these, LR outperformed the other models, achieving AUC values of 0.659 in the testing group, as shown in Fig. 5C.

Univariate and multivariate analyses of clinical features and model development

A univariate analysis was first conducted to assess live birth outcomes, with clinical variables showing a p -value < 0.05 being subsequently included in the multivariate analysis. The focus of the analysis was to compute the Odds Ratios (OR) and corresponding p -values for each variable. Following both univariate and multivariate screenings, only maternal age ($p = 0.0018$) was retained for inclusion in the clinical signature (Table 2). Utilizing ten machine learning algorithms based on maternal age, the Receiver Operating Characteristic (ROC) curve analysis yielded an AUC of 0.699 for both Logistic Regression (LR) and Naive Bayes models (Supplement 1).

Signature comparison of clinical, sixth-week, eighth-week, delta and combined model

We created a Combined model that incorporates clinical, sixth-week, eighth-week, and delta features, building on the logistic regression model's superior performance across clinical, sixth-week, eighth-week, and delta data. Both the DeltaRad and Combined models performed exceptionally well in the training cohort, with robust discriminatory power indicated by their respective AUC values of 0.883 and 0.878 (Fig. 6A). In particular, the DeltaRad model demonstrated its capacity to accurately classify and successfully identify positive samples with a sensitivity of 0.891 and an accuracy of 0.881. In a similar vein, the Combined model demonstrated a well-balanced and superior overall performance, despite slightly lower specificity when compared to DeltaRad, with a sensitivity of 0.897 and an accuracy of 0.883. With an AUC of 0.747 (Fig. 6B), the Combined model was the best model in the testing group. It identified all none live birth

Table 1 Baseline comparison between training and testing groups

Features	Total (n = 528)	Training group (n = 369)	Testing group (n = 159)	P value
Blastulation time				0.4025
Day 5	343	235	108	
Day 6	185	134	51	
Embryo stage				0.8075
3	94	67	27	
4	299	206	93	
5	124	87	37	
6	11	9	2	
ICM				0.8550
1	248	176	72	
2	237	164	73	
3	43	29	14	
TE				0.3033
1	254	181	73	
2	232	163	69	
3	42	25	17	
Fertilization method				0.9547
IVF	401	281	120	
ICSI	127	88	39	
Endometrial preparation				0.6544
mNC	168	116	52	
MS	19	15	4	
HRT	256	182	74	
GnRHa-HRT	85	56	29	
Maternal age(years)	32.00 (29.00, 35.00)	32.00 (29.00, 35.00)	31.00 (29.00, 34.00)	0.1479
BMI (kg/m ²)	21.20 (19.50, 23.20)	21.40 (19.60, 23.40)	21.00 (19.50, 22.70)	0.1844
Infertility duration (years)	3.50 (2.00, 6.00)	3.17 (2.00, 5.50)	4.00 (2.04, 6.00)	0.2806
Endometrial thickness (cm)	9.70 (8.60, 11.00)	9.70 (8.50, 10.80)	9.50 (8.60, 11.20)	0.4889
Live Birth				0.9108
Yes	472	329	143	
No	56	40	16	

ICM Inner Cell Mass, TE Trophoblast Epithelium, IVF In Vitro Fertilization, ICSI Intracytoplasmic Sperm Injection, mNC Modified natural cycle, MS Mild stimulation cycle, HRT Hormone replacement therapy, GnRHa-HRT HRT combined with a GnRH agonist, BMI Body Mass Index

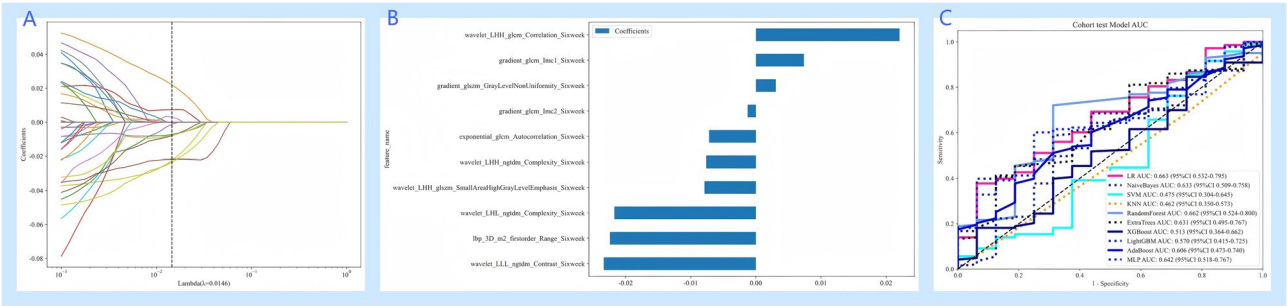


Fig. 3 Sixth week ultrasound radiomic feature screening and model construction

patients despite having a comparatively low sensitivity (0.476) and specificity (1.000)(Table 3).

We used DCA to compare the Clinical, Sixth-week, Eighth-week, Delta, and Combined Models in order to evaluate its clinical utility. Overall, positive clinical net benefits were shown by the Delta and Combined Models (Supplement 2 A,2B). Additionally, the calibration curves showed that the Delta and Combined Models both performed well in terms of calibration, with results that were consistently validated in both the testing and training cohorts (Supplement 3 A, 3B).

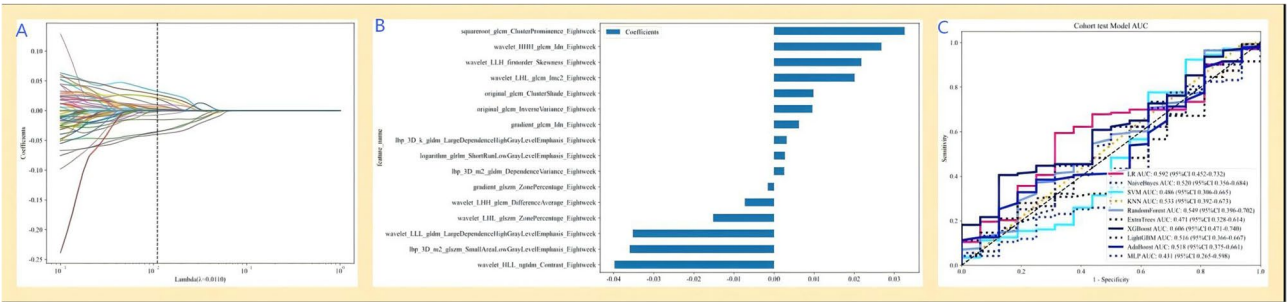


Fig. 4 Eighth week ultrasound radiomic feature screening and model construction

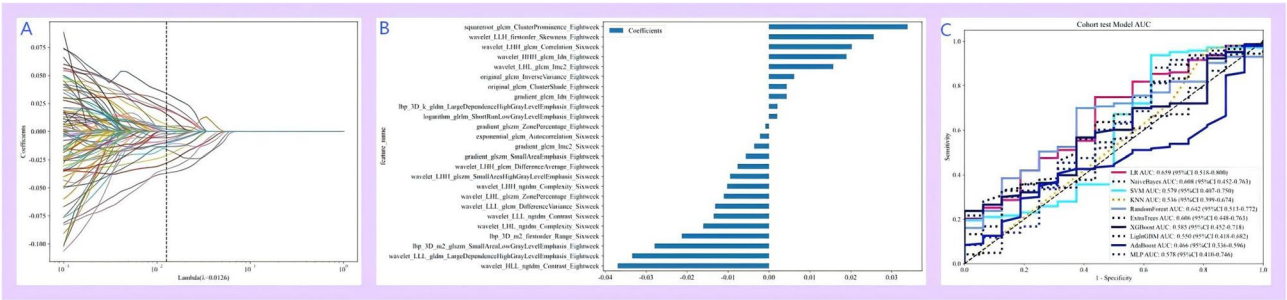


Fig. 5 Delta radiomic feature screening and model construction

Table 2 Univariate and multivariate analyses of clinical features

Features	Univariate Analysis		Multivariate Analysis	
	Odds Ratio, 95% CI	p-value	Odds Ratio, 95% CI	p-value
Blastulation time	0.84(0.43, 1.64)	0.6081		
Blastocyst Stage	0.81(0.51, 1.28)	0.3608		
ICM	0.87(0.52, 1.46)	0.6073		
TE	1.01(0.59, 1.72)	0.9806		
Maternal age (years)	0.88(0.81, 0.95)	0.0018	0.88(0.81, 0.95)	0.0018
BMI (kg/m ²)	0.96(0.86, 1.07)	0.4871		
Infertility duration (years)	0.99(0.89, 1.10)	0.8095		
Fertilization method	1.54(0.66, 3.61)	0.3215		
Endometrial preparation	0.72(0.52, 1.00)	0.0533		
Endometrial thickness (cm)	0.92(0.77, 1.11)	0.4094		

ICM Inner Cell Mass, TE Trophoblast Epithelium, BMI Body Mass Index

Development and validation of a nomogram

We selected female age and Delta radiomic score as variables to construct a nomogram. The results of the nomogram demonstrated a negative correlation between female age and live birth outcome, whereas the Delta radiomic score exhibited a positive correlation with live birth outcome(Fig. 7).

SHAP explanation of delta model

In the SHAP interpretation, the Beeswarm plot illustrates the importance of the 25 ultrasound features included in the Delta_Radiomics model for predicting live birth outcomes, highlighting the contribution of each feature to the prediction. This plot provides insight into the impact of each individual feature on the prediction of live birth

outcomes(Fig. 8A). Additionally, a waterfall plot of a representative case is provided to further demonstrate the model's interpretability(Fig. 8B).

Discussion

Delta radiomics and combined model: enhancing predictive accuracy in ART

Research on ultrasound radiomics in reproductive medicine has primarily focused on the endometrium. One study has developed models combining endometrial ultrasound radiomics with clinical features to predict clinical pregnancy outcomes after FET [20]. However, there is no research in ultrasound radiomics that specifically addresses changes in the gestational sac.Our study,

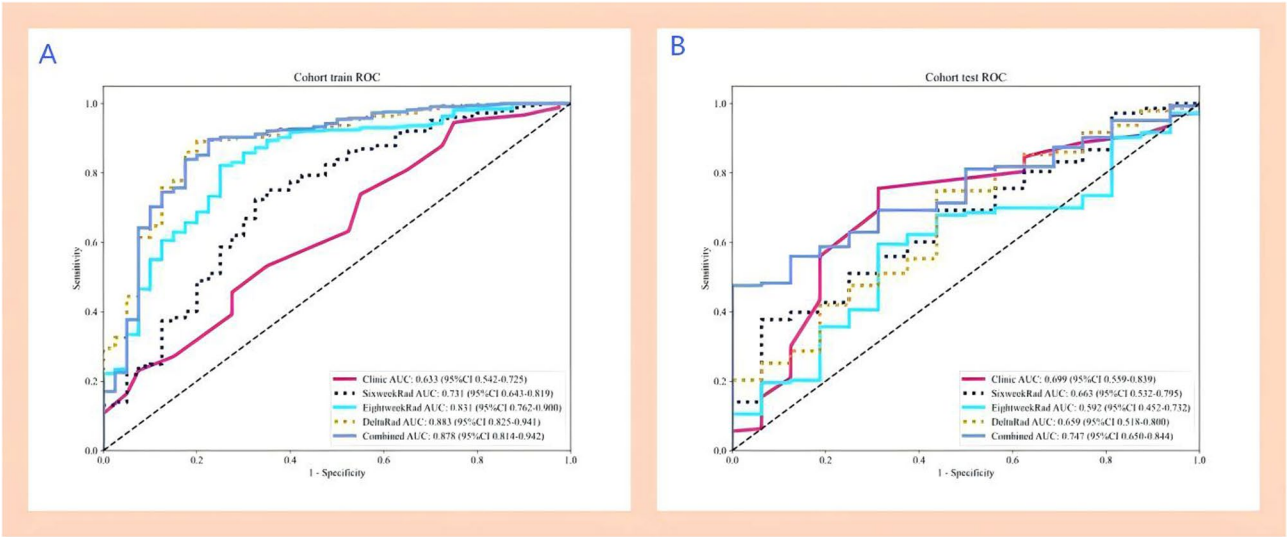


Fig. 6 Signature comparison of clinical, sixth-week, eighth-week, delta and combined model

Table 3 Signature comparison of clinical, six-week, eight-week, delta and combined model

Signature	AUC	95% CI	Sensitivity	Specificity	PPV	NPV	Accuracy	Cohort
Clinic	0.633	0.542–0.725	0.945	0.250	0.912	0.357	0.870	train
Sixth-week Rad	0.731	0.643–0.819	0.751	0.650	0.946	0.241	0.740	train
Eighth-week Rad	0.831	0.762–0.900	0.821	0.750	0.964	0.337	0.813	train
Delta Rad	0.883	0.825–0.941	0.891	0.800	0.973	0.471	0.881	train
Combined	0.878	0.814–0.942	0.897	0.775	0.970	0.477	0.883	train
Clinic	0.699	0.559–0.839	0.755	0.687	0.956	0.239	0.748	test
Sixth-week Rad	0.663	0.532–0.795	0.378	0.937	0.982	0.144	0.434	test
Eighth-week Rad	0.592	0.452–0.732	0.594	0.687	0.944	0.159	0.604	test
Delta Rad	0.660	0.518–0.800	0.748	0.562	0.939	0.200	0.730	test
Combined	0.747	0.650–0.844	0.476	1.000	1.000	0.176	0.528	test

however, focuses on the changes in the gestational sac during the sixth and eighth weeks of pregnancy.

A key finding of this study is that the predictive accuracy of the delta radiomics model improves when combining ultrasound features from both the sixth and eighth weeks of gestation with clinical data. By incorporating both temporal data and clinical information, the model can more comprehensively reflect the progression of pregnancy, which is crucial for assessing the likelihood of live birth outcomes. Single-time-point models may fail to capture the dynamic changes that occur during early pregnancy. Early developmental ultrasound images of the gestational sac can reveal critical details about pregnancy viability, which evolve over time. The delta radiomics model, by integrating features from both the sixth and eighth weeks along with clinical data, provides a more comprehensive and accurate assessment of the pregnancy’s status.

In our study, in the training cohort, both the delta radiomics model and the Combined model outperform the single models based solely on ultrasound features from either the sixth or eighth week, as well as clinical

characteristics. In the test cohort, however, the Combined model demonstrates superior performance.

Model interpretability: the role of SHAP

One major issue with machine learning models is their lack of interpretability, even though they have shown good accuracy in predicting clinical outcomes, especially when based on radiomic features [21]. Clinicians are often reluctant to rely on “black-box” models [22]. To mitigate this issue, this study incorporates SHAP values to elucidate the contributions of individual features within the delta radiomics model. SHAP, which has its roots in cooperative game theory, measures the marginal influence of each feature on the model’s predictions, offering important new information about how each feature affects the outcome.

The application of SHAP in this study has enabled a deeper understanding of the specific ultrasound features that most significantly influence the prediction of live birth outcomes. The Beeswarm plot produced in the SHAP analysis underscores the significance of 25

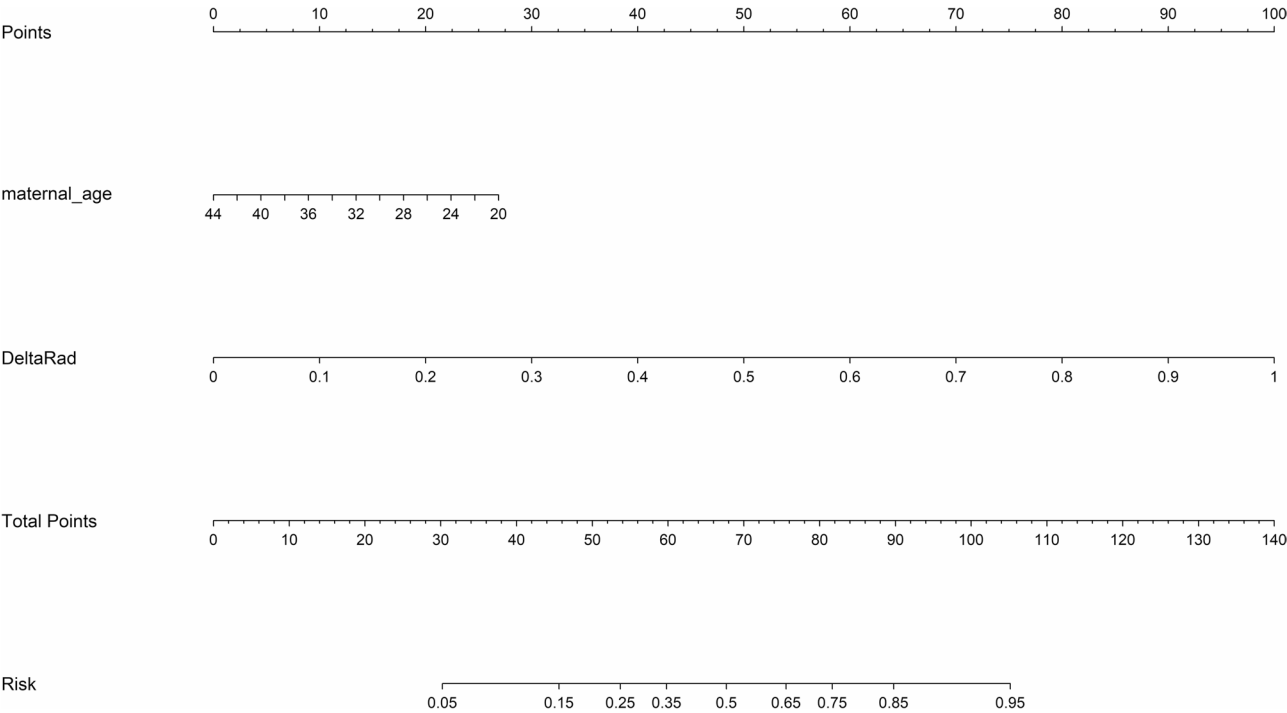


Fig. 7 Nomogram analysis of female age and delta radiomic score

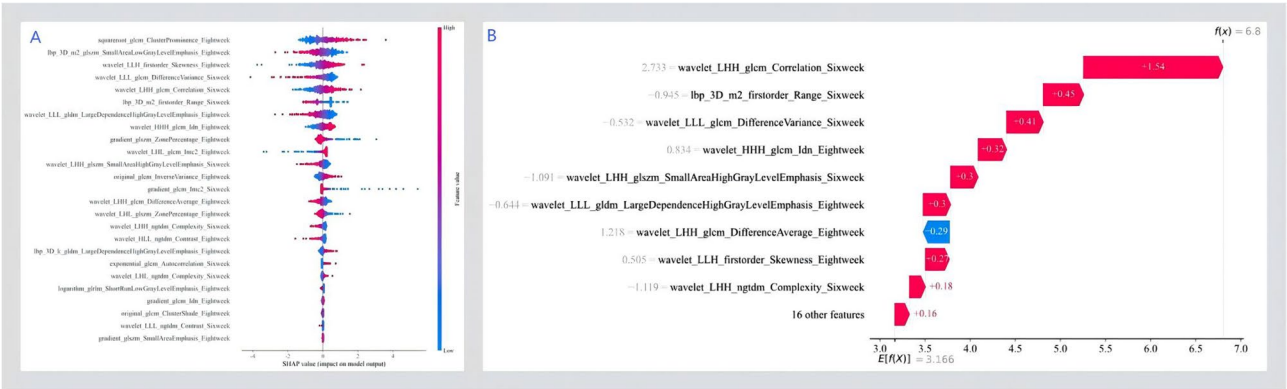


Fig. 8 SHAP explanation of delta model

ultrasound features selected from both the sixth and eighth weeks of pregnancy.

The use of a Waterfall plot for a representative case, in addition to the Beeswarm plot, helps to further clarify the prediction process on a personal level by showing how the 25 ultrasound features gradually contribute to the final prediction. Clinicians can better grasp the reasoning behind the predictions thanks to the model's increased transparency brought about by the ability to visually interpret each feature's function. In clinical settings, where openness can increase model trust and guide better decision-making, interpretability is essential.

The integration of SHAP into the radiomics-based model holds several important implications for clinical practice. First, it significantly enhances the model's

transparency, a critical factor in ensuring clinician acceptance. Second, it facilitates the identification of key features that could serve as targets for potential interventions, thereby offering actionable insights to improve clinical outcomes.

Clinical implications

In ART, the ability to predict live birth outcomes is essential. The model developed in this study, which combines clinical features and delta radiomic features, may increase the precision of these predictions. The integration of radiomic features obtained from ultrasound, such as those from both sixth-week and eighth-week gestational scans, offers a more comprehensive and dynamic understanding of early pregnancy development. If this

approach aids in identifying patients who are more likely to have live birth outcomes, clinicians may be able to intervene more effectively.

The predictions are more transparent when SHAP is used for model interpretation, which helps clinicians comprehend and have confidence in the findings. This is important when it comes to ART, since treatment choices are frequently complicated and call for careful evaluation of a number of variables.

Limitations and future directions

Although the study's findings are encouraging, it is important to recognize a number of limitations. First, because the study was conducted retrospectively, the results must first be confirmed in prospective cohorts before being widely applied in clinical practice. Second, the study only used data from one hospital, which might have limited how broadly the findings can be applied. Replicating these results across several facilities with various patient demographics should be the main goal of future studies. Third, While the clinical consistency between the training and testing cohorts has minimized the potential impact of sample size imbalance on the results, we did not apply specific statistical techniques such as weighting, stratification, or resampling to address this issue. Sample size imbalance may still contribute to selective bias, and as such, we acknowledge the need for future studies with larger, more balanced cohorts to further validate the stability and generalizability of the findings.

Conclusion

In conclusion, this study demonstrates the potential of delta radiomics and combined model, combined with SHAP-based interpretability, to enhance the prediction of live birth outcomes in assisted reproductive technology (ART). By integrating ultrasound-derived features from sixth week and eighth week along with clinical data, the delta radiomics model offers a more comprehensive and dynamic approach to early pregnancy prediction. The incorporation of SHAP further enhances the model's transparency, making it a valuable tool for clinicians in guiding treatment decisions and patient counseling. Future research should focus on validating these findings in larger, multi-center cohorts and exploring the integration of other imaging modalities to further improve predictive accuracy.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13048-025-01859-0>.

Supplementary Material 1. Univariate and Multivariate Analyses of Clinical Features and Model Development.

Supplementary Material 2. Decision Curve Analysis

Supplementary Material 3. Calibration curves Analysis

Authors' contributions

Lidan Liu and Huimei Wu contributed equally to this work as co-first authors. Lidan Liu conceived the study, designed the methodology, and drafted the manuscript. Lang Qin performed data analysis and model development. Qianyi Huang and Dongliu Xu provided ultrasound expertise and radiomics feature extraction. Mujun Li assisted with statistical analysis and interpretation. Huimei Wu supervised the research, provided clinical expertise, and critically revised the manuscript. All authors read and approved the final manuscript.

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Data availability

The data from the study will be shared upon reasonable request.

Declarations

Ethics approval and consent to participate

This retrospective study was approved by the Ethics Committee of the First Affiliated Hospital of Guangxi Medical University on July 14, 2025 (Approval No.: 2025-E0580). The requirement for informed consent was waived as the study utilized anonymized early pregnancy ultrasound images and archived medical records. All procedures complied with the ethical standards of the institutional review board and the Declaration of Helsinki (2013 revision).

AI Assistance Statement

AI Assistance Statement: AI tools (ChatGPT4o) were utilized to enhance language fluency and readability. The authors are entirely responsible for the research integrity, findings, and conclusions of this work.

Competing interests

The authors declare no competing interests.

Conflict of interest

The authors declare no competing financial interests or personal relationships that could influence the work reported in this paper.

Attestation Statement

- 1) The trial is not a clinical trial.
- 2) Data regarding any of the subjects in the study has not been previously published unless specified.
- 3) Data will be made available to the editors of the journal for review or inquiry upon request.

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